How Much Do High Schools Matter?
Causal Effects on Achievement and Major Choice

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Motivation

Question

- How much do high schools matter for students' later academic achievement and higher-education major choice?
- The economics literature has strong evidence on school effects on test scores, but much less on whether schools affect *field of study*. (Angrist et al. 2016; Dobbie and Fryer 2015; Angrist et al. 2017; Angrist et al. 2024)
- One empirical challenge is that "school attended" is a high-dimensional treatment: one treatment for each high school.

Original Problem: Many Treatments

- My original goal was to estimate idiosyncratic effects of each school.
- A fully flexible treatment would include one indicator for each high school:

$$D_i^{\text{full}} = \left(\mathbf{1}\{S_i = 1\}, \mathbf{1}\{S_i = 2\}, \ldots, \mathbf{1}\{S_i = J\}\right).$$

- The lottery shifts students across schools, but generally does not identify a separate causal effect for every school.

Original Problem: Common Fallback

- Suppose all students rank three schools $A \succ B \succ C$.
- Suppose they all have the same guaranteed fallback school G.
- Then lottery-induced comparisons can be interpreted relative to the same counterfactual:

$$\beta_{AG}, \qquad \beta_{BG}, \qquad \beta_{CG}.$$

- Because the comparison school is common, these objects are comparable across target schools.

Original Problem: Different Fallbacks

- In practice, students have different fallback schools:

$$G_i \in \{G^1, G^2, \ldots, G^K\}.$$

- The relevant objects become target-school-by-fallback effects:

$$\beta_{AG^1}, \beta_{AG^2}, \ldots, \beta_{CG^K}.$$

- A gain from G^1 to A is not automatically comparable to a gain from G^7 to B.

- This motivates projecting schools into a scalar value index instead of trying to recover every pairwise school contrast.

Collapsing to meaningful margins: Kirkeboen et al. (2016)

- Kirkeboen et al. (2016) study field-of-study effects using centralized assignment in Norway.
- Because applicants have different next-best alternatives, comparing each program would put us in a similar situation than the one described before.
- Their solution: collapse programs into broad fields and estimate next-best effects across fields.
- The cost of this approach is that it requires ignoring effects(variation) coming from differences in institution in their main specification.

Collapsing to meaningful margins: Next-Best Effects

Table 5. 2SLS estimates of the payoffs to field of study (USD 1,000)

	Next best alternative (k):								
	Humanities	Soc Science	Teaching	Health	Science	Engineering	Technology	Business	Law
Completed field (j):									
Humanities		21.38*	-4.72	-22.93*	4.97	-38.51**	6.87	-42.21**	-156.33
		(10.97)	(9.85)	(12.12)	(11.86)	(14.72)	(48.29)	(10.56)	(437.28)
Social Science	18.72**		9.84	-10.82	55.46**	-55.36**	-110.38	-28.37**	-76.07
	(6.73)		(11.55)	(13.00)	(21.45)	(20.60)	(102.97)	(10.66)	(86.42)
Teaching	22.25**	31.37**		1.82	23.46**	-33.94**	-35.32	-21.08**	22.78
	(4.96)	(7.88)		(6.55)	(9.45)	(12.54)	(37.07)	(7.12)	(127.87)
Health	18.75**	30.69**	7.72**		28.87**	-27.87**	-43.38**	-17.39**	-55.19
	(6.25)	(7.56)	(2.82)		(7.64)	(10.35)	(20.84)	(3.97)	(97.68)
Science	53.71**	69.59**	38.58**	29.63**		-2.21	16.81	-4.92	148.26
	(18.37)	(22.36)	(14.20)	(11.53)		(14.60)	(18.07)	(10.51)	(276.20)
Engineering	59.81	-5.53	75.24**	0.16	52.35**		-46.00	-13.03	-57.66
	(50.59)	(58.17)	(37.50)	(16.36)	(20.98)		(43.89)	(23.70)	(166.60)
Technology	41.87**	58.69**	22.08*	32.45**	68.07**	-5.56		7.03	-53.07
	(10.84)	(10.09)	(12.44)	(10.09)	(9.63)	(11.95)		(9.49)	(147.53)
Business	48.13**	61.93**	31.02**	30.22**	58.01**	-3.42	28.54*		3.53
	(11.25)	(12.03)	(8.78)	(10.86)	(10.48)	(12.61)	(15.61)		(83.04)
Law	46.34**	55.62**	36.60**	21.49*	40.07**	-27.53	-15.55	-1.36	
	(7.16)	(8.34)	(11.56)	(11.46)	(9.68)	(18.29)	(17.96)	(8.66)	
Medicine	83.34**	79.39**	62.62**	45.57**	81.31**	21.07	40.07**	23.34**	14.82
	(9.76)	(10.65)	(9.02)	(7.01)	(9.71)	(20.67)	(11.72)	(8.79)	(83.61)
Average y^k	30.01	23.40	46.15	51.79	27.31	87.85	78.37	75.61	105.83
Observations	8,391	11,030	10,987	3,269	6,422	3,085	1,245	4,403	1,251

Note: From 2SLS estimation of equations (14)-(15), we obtain a matrix of the payoffs to field j as compared to k for those who prefer j and have k as next-best field. Each cell is a 2SLS estimate (with st. errors in parenthesis) of the payoff to a given pair of preferred field and next-best field. The rows represent completed fields and the columns represent next-best fields. The row labeled average y^k reports the weighted average of the levels of potential earnings for compliers in the given next-best field. The final row reports the number of observations for every next-best field. Stars indicate statistical significance, * 0.10, ** 0.05.

Source: (Kirkeboen et al. 2016).

Main Pivot

- Instead of estimating one causal effect for each school, collapse school attendance into a scalar school-value treatment.
- For each school s, construct an observational school value β_s.
- A student's treatment is the value of the school they attend:

$$D_i = \beta_{S_i}.$$

- The IV asks whether lottery-induced movement toward higher-value schools (in a given metric) causally improves outcomes.
- Since the variation comes from lottery offers, the effects tells us about the causal effects of schools. But at the same time, these results are mediated by a certain projection, which might nor not be relevant.

Data and School Values

Data

- Universe: students observed in regular grade 8 in Chile between 2017 and 2020.
- Approx. 945,000 unique students.
- Linked records:
  - grade-4 SIMCE math and language scores;
  - grade-8 cohort, gender, age, and home comuna;
  - school enrollment histories after grade 8;
  - PSU/PAES math and language scores;
  - higher-education enrollment and STEM field indicators.

Assigning a High School

- For each student, I track school enrollment during the four years after grade 8.
- Main school assignment:

$$S_i = \text{school code attended for the most post-grade-8 years.}$$

- Ties are broken toward the most recent school.
- This maps each student in the broad grade-8 universe to one high-school code when available.

Raw and Adjusted School Values

Raw school mean

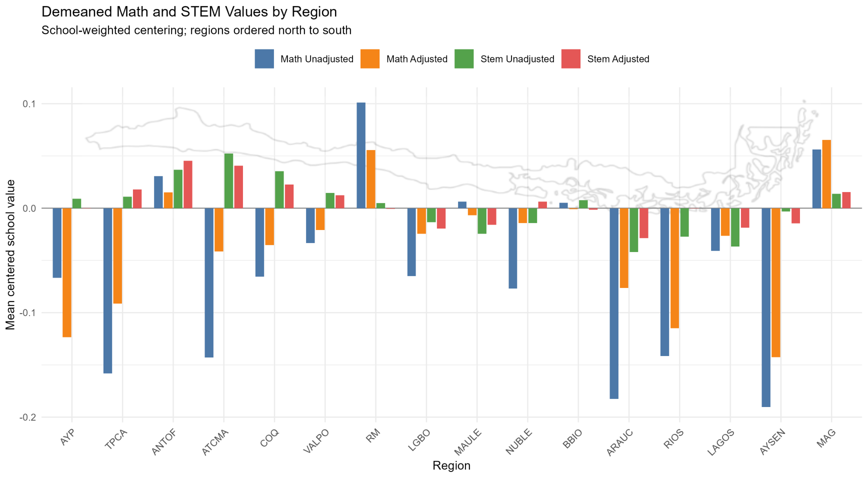
$$\beta_s^{\text{raw}} = \frac{1}{N_s} \sum_{i:S_i=s} Y_i.$$

Adjusted observational value-added

$$Y_i = \mu + \beta_{S_i} + \sum_{p=1}^{3} \lambda_{Mp} M_i^p + \sum_{p=1}^{3} \lambda_{Lp} L_i^p + \gamma_{\text{cohort}(i)} + \delta_{\text{gender}(i)} + \rho_{\text{age}(i)} + \kappa_{\text{comuna}(i)} + \varepsilon_i.$$

- β_s is the school fixed effect. I use this as an observational, selection-contaminated measure of school value. (Chetty et al. 2014; Angrist et al. 2017)
- For these results to be causal, we would need to assume that the controls remove all selection on unobservables that is correlated with the school attended. *(CIA)*
- I have also a noisy measure of family income (G4), attendance and performance within schools (G 1-8). Some signal there but skeptical.

Adjustment Matters and Patterns Are Sensible



IV Design

Lottery Sample

- I use students who enter the grade-9 centralized assignment mechanism *(SAE)* on time (i.e. same year they are observed in grade 8).

- I restrict to students with non-trivial assignment risk:

$$\max_{s \in \mathcal{S}} p_{is} < 1,$$

  where p_{is} is the simulated prob. that student i is assigned to school s.

- This implies going from $\approx$ 320,000 students that enter SAE, to $\approx$ 165,000 students with assignment risk. This is roughly 1/6 of students in the universe having assignment risk.

- The lottery offer provides exogenous variation in the value of the school attended, conditional on assignment-probability controls.

Supported School Set

- Full probability controls are high-dimensional. Concretely 2 per school: one for the probability and one for the zero-probability indicator.
- I define the supported lottery school set as schools with positive assignment probability for at least 100 at-risk students:

$$\mathcal{S}^{100} = \left\{ s : \sum_i \mathbf{1}\{p_{is} > 0\} \geq 100 \right\}.$$

- This keeps the current design computationally feasible while covering a large share of students.

Support Tradeoff

Threshold students at risk	School group	Schools	Students	Share
100	SAE $<$ 100	1,491	315,261	33.4%
100	SAE $\geq$ 100	1,011	493,955	52.3%
100	Particular Pagado	497	87,373	9.3%
100	Other non-SAE schools	1,134	47,771	5.1%
50	SAE $<$ 50	940	174,922	18.5%
50	SAE $\geq$ 50	1,562	634,294	67.2%
50	Particular Pagado	497	87,373	9.3%
50	Other non-SAE schools	1,134	47,771	5.1%
25	SAE $<$ 25	530	89,266	9.5%
25	SAE $\geq$ 25	1,972	719,950	76.2%
25	Particular Pagado	497	87,373	9.3%
25	Other non-SAE schools	1,134	47,771	5.1%

Using the Value-Added Metric: Treatment and Instrument

The endogenous school-value treatment is

$$D_i = \beta_{S_i}.$$

The excluded instrument is

$$Z_i = \mathbf{1}\{O_i \in \mathcal{S}^{100}\}\beta_{O_i}.$$

- S_i: school attended after grade 8.
- O_i: first-round lottery offer.
- Offers outside $\mathcal{S}^{100}$ are assigned zero in the excluded instrument.

2SLS System

$$D_i = \pi_0 + \pi_1 Z_i + P_i' \psi_D + X_i' \lambda_D + u_i.$$

$$Y_i = \alpha + \theta \widehat{D}_i + P_i' \psi_Y + X_i' \lambda_Y + \varepsilon_i.$$

- P_i: assignment-probability controls for schools in $\mathcal{S}^{100}$, including probabilities and zero-probability indicators.
- X_i: grade-8 cohort, gender, grade-8 age, fourth-grade math score, and fourth-grade language score. Probably moving to higher-order polynomials.
- θ: causal return to a lottery-induced increase in the scalar school-value index.

Preliminary Results

How to Read θ

- The general interpretation is that a student induced by a lottery offer to attend a school (a complier) with a 1-unit higher value in the respective metric is expected to have a θ-unit higher outcome.

- Here, θ compares causal gains to the observational school-value metric. If $\theta = 1$, then a one-unit difference in observational school value maps one-for-one into a causal effect. This idea is formalized and a test for it is proposed in Angrist et al. (2017).

Main Results

	Math VA		Language VA		STEM VA	
	Unadj.	Adj.	Unadj.	Adj.	Unadj.	Adj.
θ	0.208	0.431	0.167	0.393	0.345	0.584
SE	0.033	0.057	0.032	0.062	0.046	0.068
First-stage coef.	0.455	0.457	0.439	0.435	0.471	0.455
First-stage F	4399	3656	5481	4899	6509	5086
N	105,902	105,893	105,902	105,894	135,588	135,588

All p-values are below 0.001.

Interpretation

- VA metrics are predictive of lottery-induced causal effects across specifications.
- Results are metric-specific, but also show that exogenous changes in school assignments do change outcomes for individuals.
- To my knowledge, results are among the first to show that school assignment lotteries induce changes in higher education major choice, which is an important contribution in itself.

STEM Enrollment by Gender

	STEM VA			STEM gender gap reduc. VA		
	All	Boys	Girls	All	Boys	Girls
θ	0.584	0.535	0.695	-0.254	-0.477	0.000
SE	0.068	0.088	0.097	0.078	0.122	0.087
p-value	< 0.001	< 0.001	< 0.001	0.001	< 0.001	0.998
First-stage coef.	0.455	0.460	0.443	0.443	0.457	0.429
First-stage F	5086	3198	2003	4954	2810	2154
N	135,588	67,221	68,367	125,672	63,669	62,003

STEM Gender Interpretation

- Schools with higher adjusted STEM VA raise STEM enrollment for both boys and girls.
- The point estimate is larger for girls than boys, but the difference is not statistically significant in the current separate-sample comparison.
- The gender-gap-reduction VA is defined as female minus male STEM value-added.
- Preliminary evidence suggests that schools with higher gender-gap-reduction VA reduce the male-favored gap mainly through lower boys' STEM enrollment, rather than clear increases for girls.

School Funding Per Person

- I also apply the same scalar-treatment framework to school resources.
- For each school, I construct public funding per student from MINEDUC subsidy payments to schools.
- I use two school-level measures:
  - average public funding per student over 2017–2021;
  - growth in public funding per student from 2017–2018 to 2019–2021.
- Both measures are expressed in millions of 2021 pesos per student, roughly thousands of 2021 USD per student.

Funding Per Person Results

	Funding per student			Growth in funding per student		
	Math	Language	STEM	Math	Language	STEM
θ	-0.022	-0.041	0.029	-0.020	-0.001	0.031
SE	0.027	0.028	0.014	0.048	0.050	0.021
p-value	0.416	0.144	0.042	0.685	0.979	0.132
First-stage coef.	0.339	0.339	0.268	0.398	0.398	0.343
First-stage F	73.6	73.6	53.4	139.7	139.7	127.7
N	97,303	97,303	123,953	96,501	96,501	123,052

Treatments are measured in millions of 2021 pesos per student.

Funding Interpretation

- The expenditure-based scalar metric is not consistently predictive of lottery-induced gains in test scores or STEM enrollment.

- The first stages are strong, so the null-ish reduced relationship is not coming from a weak instrument. However, the first-stage is much weaker than in the previous specifications.

- **What these metrics capture matters for the results:** public funding for these schools is largely determined by enrollment and thus should be constant per student within schools.

- The non-enrolmment component of funding tends to target specific student populations, so higher funding per student is not automatically a pure resource shock.

Takeaways

- The scalar school-value approach makes the lottery variation tractable without estimating one effect per school.
- Observational school VA predicts lottery-induced causal gains in achievement and STEM enrollment.
- Adjusted VA performs better than raw means.
- The current design suggests schools affect higher-education major choice, not only test scores.

Next Short-Term Steps

- Further clean, impute and include family income in the VA metrics.
- Explore run-time of the 50- and 25-student support thresholds.
- Test gender differences using pooled interaction specifications.

Extension: Do students moved into STEM programs actually do better?

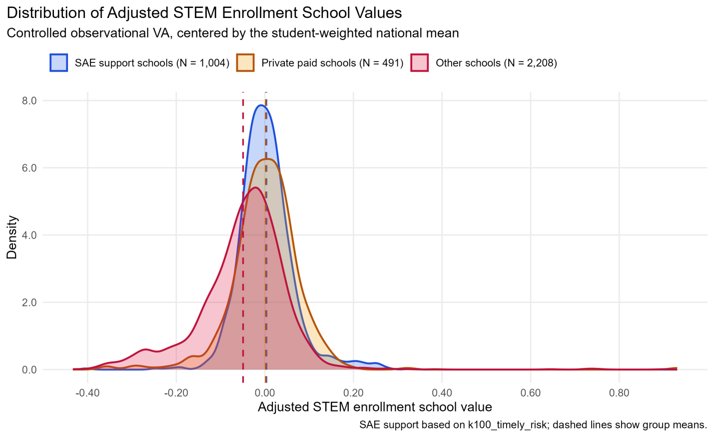
- Discretize the STEM VA metric and characterize compliers using Abadie (2003).

- If I had data, I can run a regression of wages on STEM enrollment –and controls– for the (school-lottery) compliers above i.e. using the κ weights from the Abadie (2003) characterization.

- This might be possible in 5 years or so since these individuals are just starting college. An alternative is to look at intermediate outcomes like college grades or persistence in STEM majors.

- Another alternative is to look at similar populations in STEM programs in the past, under a version of Conditional Effects Ignorability (CEI) (Angrist and Fernandez-Val 2010) .

Distribution of STEM School VA



Thanks!